

Geometry and material data

slot profile, $B(H)$ 81 points, magnet data
loss coefficients — read from the FE project

Mesh

M_s columns $\times$ L_s layers (Fig. 3)
four branches per cell (Fig. 4)

Air-gap operator

harmonic assembly, $N = \lfloor M_s/2 \rfloor$
tiling + condensation if doubly slotted (Fig. 5)

computed **once** per machine

Newton on the iron

closed-form B inversion, Jacobian R_d
line search on the nodal residual

Convergence

relative nodal residual $< 10^{-9}$,
not stagnation of B

Post-processing

B_r , B_t in closed form; torque by Parseval sum
iron and magnet losses; flux linkage

not converged

next rotor position